THEJUS MAHAJAN PH.D.

Bioinformatics & Data Scientist

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Google Scholar

CAREER INTERESTS

Computational Scientist (PhD) with 5+ years of experience in **large-scale data analysis**, **statistical modeling**, and **pipeline development**. Trained in **R (Bioconductor, Tidymodels)** and **Python (Pandas, NumPy, Scikit-learn)** for high-dimensional data analysis. Built a viral genomics pipeline using **Nextflow (DSL2)**. Currently deepening skills in data pipeline development, **genomics**, AI/machine learning, and NGS analysis through specialized training and an industry internship. Eager to develop expertise in **single-cell multi-omics** and apply computational and data science skills to advance research in **stem cell biology and genomics**.

EXPERIENCE

Internship – Bioinformatics & Data Pipeline Engineering
HealthTwiSt GmbH
02/2026 - 04/2026 | Berlin, Germany

Clinical data research and development under Dr. Andreas Busjahn
- Refactored a 1,834-line medical data pipeline (R/tidyverse) for Germany's interventional radiology quality registry (DeGIR); 143K rows × 1,920 columns.
- Externalized 257 hardcoded rules into 8 config files; 26.5% code reduction with byte-level output verification (`identical()`).
- Identified 2 latent bugs through systematic code analysis; documented without altering pipeline behavior.
- Machine Learning workflows using the Tidymodels framework in R.

Further Training – Bioinformatics and Biostatistics
CQ Beratung + Bildung GmbH
08/2025 - 02/2026 | Berlin, Germany

Application-related Bioinformatics and Biostatistics
- Programming methodology: Shell (BASH), Python (object-oriented), R.
- Bioinformatics databases (NCBI, Biopython, SQL) and sequence analysis.
- NGS analysis pipelines (Nextflow, Galaxy); structural bioinformatics.
- Clinical biostatistics (R/Bioconductor, ANOVA, PCA, HCA, survival analysis); GDPR/BDSG compliance.

Guest Scientist
Helmholtz-Zentrum Hereon, Ökosystemmodellierung
05/2025 - 10/2025 | Geesthacht, Germany

Computational Biologist & Systems Modeler
- Spearheaded research modeling evolutionary processes in marine ecosystems.
- Applied advanced computational techniques to simulate long-term environmental impacts.

Job Search & German Language Learning
Hamburg, Germany
02/2025 - 04/2025 | Hamburg, Germany

Active job search and German language acquisition (Goethe B1 certification).

Post-doctoral Scientist
University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences
08/2021 - 01/2025 | Hamburg, Germany

Marine Ecosystem Modelling

STRENGTHS

Data Pipeline Engineering
Refactored a production medical data pipeline (1,834 lines, 143K rows), achieving 26.5% code reduction with zero behavioral regression. Byte-level verification after every change.

Complex Problem-Solving
Resolved a critical data-loss error by creating a Fortran simulation to correct for a misplaced particle detector, recalculating results using multi-sensor data.

Bioinformatics & Biostatistics
Bridged physical science expertise with modern biological workflows. Skilled in complex multidimensional array operations, JAX/GPU acceleration, sequence alignment, and multivariate statistical modeling using R (Bioconductor) and Python.

TECHNICAL SKILLS

Programming
Python (JAX, Pandas, NumPy, Scikit-learn, Biopython), R (Bioconductor, Tidymodels), SQL, BASH

Bioinformatics
Nextflow (DSL2), NGS Analysis, Multiple Sequence Alignment (MAFFT), Sequence Trimming (trimAl), NCBI / Entrez

Biostatistics & ML
Deep Learning Foundations (TensorFlow, PyTorch concepts via JAX), Multivariate Analysis, PCA, Clustering, Scikit-learn, Tidymodels

Data Engineering
Hardware Acceleration (TPU/GPU), Large dataset processing (HDF5, NetCDF), Pipeline development

Tools
Linux/Unix, Git, HPC clusters, Conda

TRAINING / COURSES

IBM: Machine Learning with Python; Introduction to AI
Coursera (in progress)

High Performance Computing (HPC) Training
Jülich Supercomputing **Center (JSC)**,
Forschungszentrum Jülich

MITx: 7.00x Introduction to Biology - The Secret of Life

- Model Development: Developed the novel "Cyanobacteria Life Cycle" (CLC) model within the ERGOM framework for climate warming projections.
- High-Performance Computing & ML: Led the translation of a legacy Fortran continuous model into a high-performance Python simulation engine. Utilized **Google JAX** vectorization and branchless logic for massive parallelization (direct compatibility with **TPUs and GPUs**).
- This architecture established a strong foundation in tensor operations underpinning deep learning frameworks like **PyTorch** and **TensorFlow**.

Job Search
Kerala, India
04/2021 - 07/2021 Kerala, India

Relocation from India to Germany and active job search.

Physics Educator
Tutorwaves Solutions Inc / Freelance
10/2018 - 03/2021 Kerala, India

Prepared physics content for competitive examinations and coached state-level chess teams.

Ph.D. Researcher in Astrochemistry
University of Paris-Saclay
10/2015 - 09/2018 Orsay, France

Led atomic collision experiments, analyzed data, and modeled outcomes.
- Acquired and processed terabytes of raw data from tandem particle accelerator experiments.
- C++ signal processing, algorithm development, and molecular fragmentation modeling (KIDA database).

EDUCATION

Ph.D. in Astrochemistry
Université Paris-Saclay, Institute of Molecular Sciences (ISMO) Orsay, France
09/2015 - 09/2018 Orsay, France

M.Sc. in Physics
National Institute of Technology Calicut, India
07/2012 - 01/2015 Calicut, India

B.Sc. in Physics
University of Calicut, India
06/2009 - 04/2012 Thrissur, India

Comprehensive coursework on genetics, biochemistry, and molecular biology.

REFERENCES

Dr. Andreas Busjahn
abusjahn@healthtwist.de

Prof. Dr. Kai Wirtz
kai.wirtz@hereon.de

Prof. Dr. Elisa Schaum
elisa.schaum@uni-hamburg.de

OTHER SELECTED PROJECTS

Hepatitis Delta Virus Pipeline • GitHub
Nextflow (DSL2) pipeline for viral sequence analysis: NCBI Entrez download, MAFFT alignment, trimAl trimming, Conda-managed dependencies.

JAX/TPU High-Performance Simulator • GitHub
Ported a legacy Fortran model to a modular Python architecture using Google JAX. Engineered branchless, vectorized logic for massive TPU/GPU parallelization.

Introduction to Tidymodels • GitHub
Teaching script for ML in R: recipes, parsnip, workflows, and yardstick on the Palmer Penguins dataset.

Berlin Tree Analysis • GitHub
Interactive Jupyter notebook analyzing 5 years of tree planting data in Berlin using Python, Pandas, and Seaborn.

PUBLICATIONS

5 peer-reviewed articles • Google Scholar Profile

LANGUAGES

English (4/5)
Proficient

German (3/5)
B1 - Goethe Certified

French (2/5)
Intermediate

Malayalam (5/5)
Native

Tamil (4/5)
Proficient

Hindi (2/5)
Intermediate